Unit NCIR2

SKU:U150





















Description

Unit NCIR2 is a non-contact infrared single-point temperature measurement unit equipped with a data processing MCU, featuring a temperature measurement range of -70°C to 380°C. The MCU uses STM32, and through data processing, it can achieve high and low-temperature alarm functions. The board integrates a buzzer, RGB indicator light, a function button, and a reset button, and communicates with the host via I2C. This product can be used in conjunction with a host or independently, making it suitable for temperature measurement and abnormal alarm scenarios.

Features

- STM32F030F4P6
- MLX90614 Infrared (IR) Sensor
- GROVE I2C/HY2.0-4P Interface
- Programming Platform: Arduino/UIFlow
- Two LEGO-compatible holes

Includes

- 1 x Unit NCIR2
- 1 x HY2.0-4P Grove Cable (20cm)

Applications

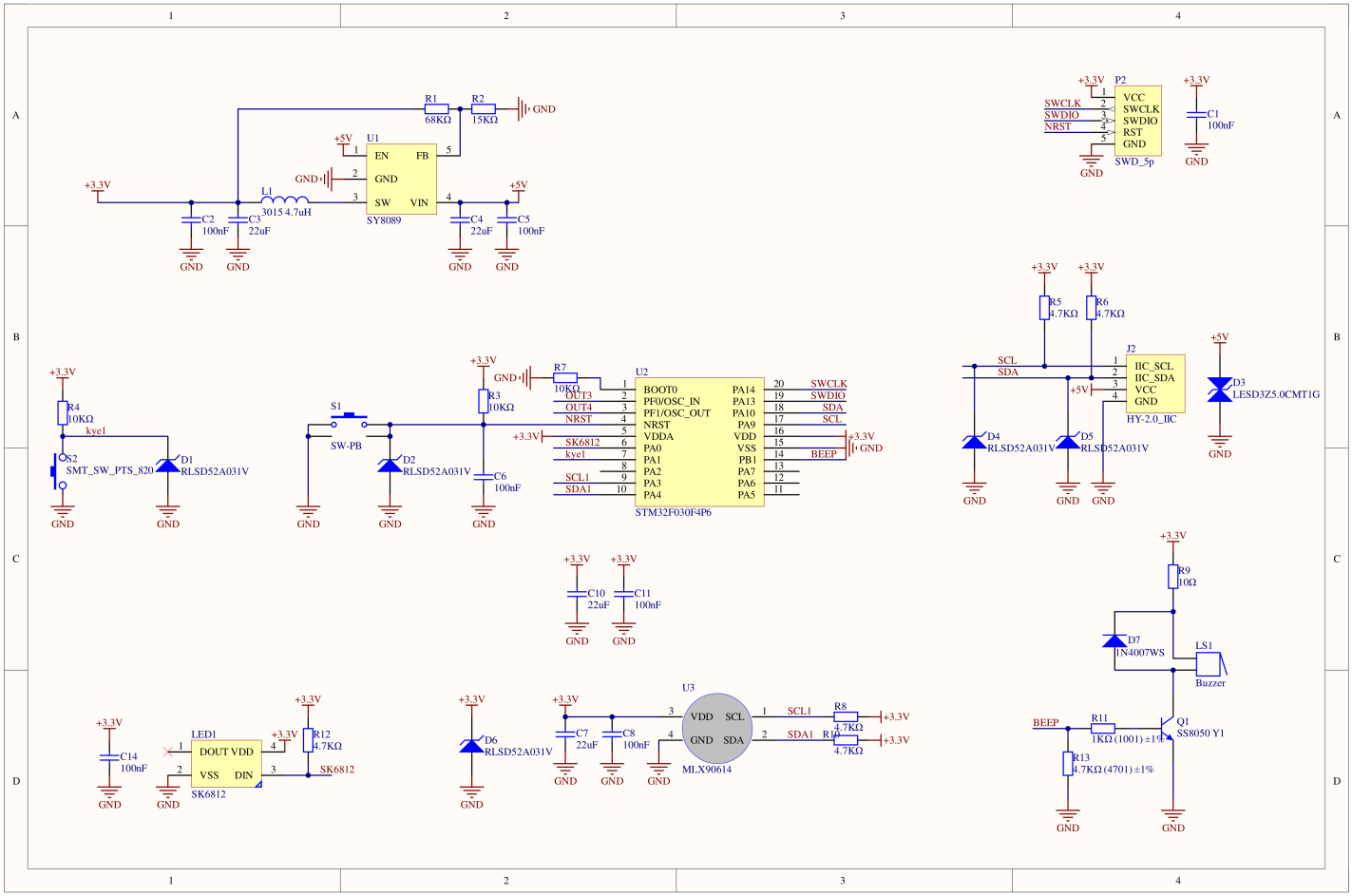
- High-precision non-contact temperature measurement
- Motion detection
- Smart home

Specifications

Specification	Parameter
MCU	STM32F030F4P6
Sensor	MLX90614ESF-BAA
Protocol	I2C @0x5A
Measured Object Temperature Range	-70°C ~ 380°C
Sensor Operating Temperature	-40°C ~ 125˚C
Unit Operating Temperature	0°C ~ 40°C
Measurement Accuracy	±0.5°C
Development Platform	Arduino, UIFlow
Product Size	48.0 x 24.0 x 8.0mm
Product Weight	7.3g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	12.8g

Schematics

- Unit NCIR2 Schematics PDF

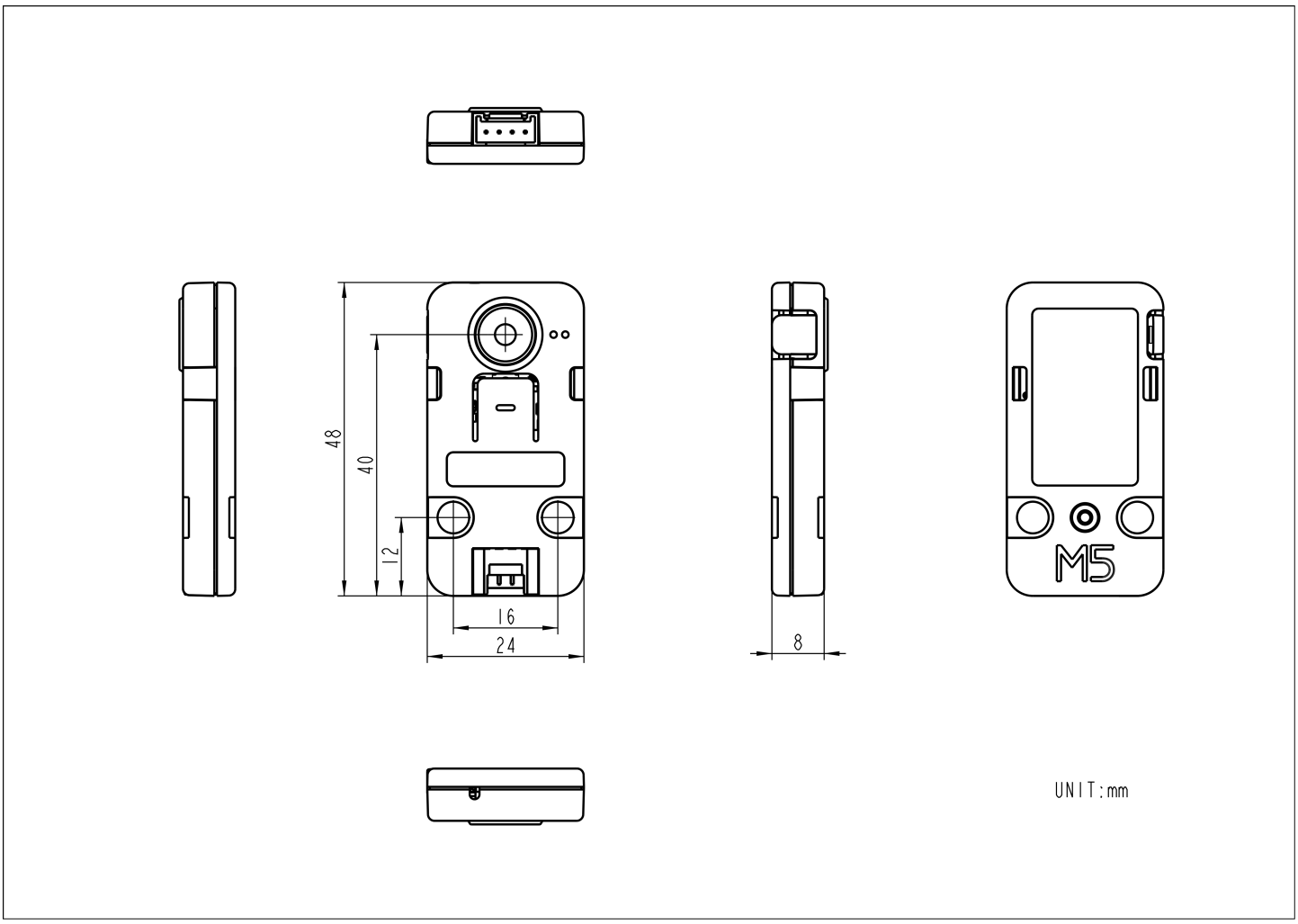


PinMap

Unit NCIR2

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit NCIR2 Structure Files

Datasheets

- STM32F030F4P6 Datasheet
- MLX90614 Datasheet

Softwares

Arduino

- Unit NCIR2 Arduino Library



UiFlow1

- Unit NCIR2 UiFlow1 Docs

UiFlow2

- Unit NCIR2 UiFlow2 Docs

Internal Firmware

- Unit NCIR2 Internal Firmware

Protocol

- Unit NCIR2 I2C Protocol

REG MAP (Addr:0x5A)		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	note
Temperature	**0x00 R**	Temperature-L	Temperature-H															**Temperature:** unit Celsius (actual temp value) * 100[1]
Emissivity	**0x10 R/W**	Emissivity-L	Emissivity-H															**Emissivity:** 65535 * rate **(0.1 < rate < 1)**[2]
Alarm Temperature	**0x20 R/W**	Alarm Low Temp-L	Alarm Low Temp-H	Alarm High Temp-L	Alarm High Temp-H													**Alarm Temp:** unit Celsius (actual temp value) * 100[3]
Alarm LED[10]	**0x30 R/W**	Alarm Low Temp-R	Alarm Low Temp-G	Alarm Low Temp-B	Alarm High Temp-R	Alarm High Temp-G	Alarm High Temp-B											**Alarm LED R/G/B:** 0~255
Alarm Buzz[11]	**0x40 R/W**	Alarm Low Temp Buzz Freq-L	Alarm Low Temp Buzz Freq-H	Alarm Low Temp Buzz Interval-L	Alarm Low Temp Buzz Interval-H	Alarm Low Temp Buzz Duty	Alarm High Temp Buzz Freq-L	Alarm High Temp Buzz Freq-H	Alarm High Temp Buzz Interval-L	Alarm High Temp Buzz Interval-H	Alarm High Temp Buzz Duty							**Alarm Temp Buzz Freq**[4]**, Alarm Temp Buzz Interval**[5]**, Alarm Temp Buzz Duty**[6]
Buzz	**0x50 R/W**	Buzz-Freq-L	Buzz-Freq-H	Buzz Duty	Buzz Control													**Buzz Freq**[7]**, Buzz Duty**[8] **Buzz Control:** 0, disable; 1, enable
RGB LED	**0x60 R/W**	LED-R	LED-G	LED-B														**LED R/G/B: 0~255**
Button	**0x70 R**	Button Status																**Button Status:** 0 or 1
Save Config	**0x80 W**	Save Config																**Save Config:** write 1 to save config
Chip Temperature	**0x90 R**	Chip Temperature-L	Chip Temperature-H															**Chip Temperature:** unit Celsius (actual temp value) * 100[9]
Firmware Version	**0xF0 R**															Version		**Version:** firmware version number
I2C Address	**0xF0 R/W**																Address	**Address:** 1~127

[1] For example, the actual temperature is 27.55 degrees Celsius, and the obtained data is 27.55*100=2755，Temperature-L = 0xC3, Temperature-H = 0x0A

[2] For example, the emissivity needs to be set to 0.95, and the value to be set is 65535*0.95=62258, Emissivity-L = 0x32, Emissivity-H = 0xF3.(Note: Everytime you set the emissivity, must reconnect the NCIR2)

[3] For example, to set the high temperature alarm temperature to 37.00 degrees Celsius, the value to be set is 37.00*100=3700， Alarm Low Temp-L = 0x74, Alarm Low Temp-H= 0x0E

[4] Alarm Temp Buzz Freq: The unit is Hz. For example, when setting the high temperature alarm, the Buzz frequency is 4000Hz, Alarm Low Temp Buzz Freq-L = 0xA0, Alarm Low Temp Buzz Freq-H = 0x0F

[5] Alarm Temp Buzz Interval: The unit is ms. For example, set the Buzz interval to 100ms, Alarm Low Temp Buzz Freq-L = 0x64, Alarm Low Temp Buzz Freq-H = 0.(Note: The value is from 1ms to 5000ms)

[6] Alarm Temp Buzz duty. For example, set the Buzz duty to 50%, Alarm Low Temp Buzz Duty = 255 * 0.5 = 127

[7] Buzz Freq: The unit is Hz. For example, set the buzz frequency to 4000Hz, Buzz-Freq-L = 0xA0, Buzz-Freq-H = 0x0F

[8] Alarm Temp Buzz duty. For example, set Buzz duty to 50%, Buzz Duty = 255 * 0.5 = 127

[9] For example, the actual temperature is 27.55 degrees Celsius, and the obtained data is 27.55*100=2755，Chip Temperature-L = 0xC3, Chip Temperature-H = 0x0A

[10] If you want to disable alarm led, set Alarm Low Temp-R=0, set Alarm Low Temp-G=0, Alarm High Temp-B=0, Alarm High Temp-R=0, set Alarm High Temp-G=0, Alarm High Temp-B=0

[11] If you want to disable alarm buzz, set Alarm Low Temp Buzz Duty=0, set Alarm High Temp Buzz Duty=0

Video